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DSA0604 Data Handling and Visualization-Slot B

CO1 – Assessment Test 2 (AT2)
Data Interpretation Exercise
Instructions:

For each question, write the program in the specified language (R or Python), generate the required
visualization, and answer the interpretation sub-questions clearly with justification.

Question 1: Nonlinear Axes [Python]

Dataset

Dataset

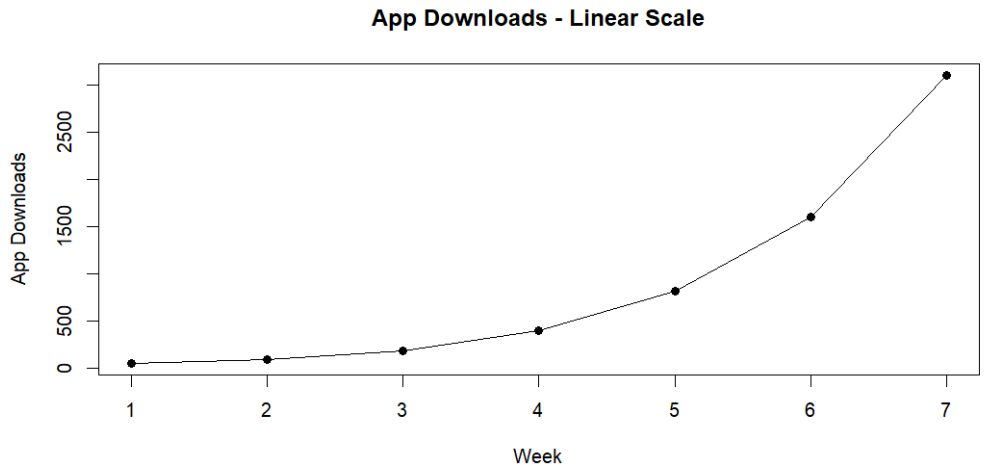
Week	App Downloads
1	50
2	95
3	180
4	400
5	820
6	1600
7	3100

Questions

1. Write a Python program to display the data using a linear y-axis.

```
# Data
week <- c(1, 2, 3, 4, 5, 6, 7)
downloads <- c(50, 95, 180, 400, 820, 1600, 3100)
# Linear scale plot
plot(week, downloads,
    type = "o",
    main = "App Downloads - Linear Scale",
    xlab = "Week",
    ylab = "App Downloads",
    pch = 16)
```

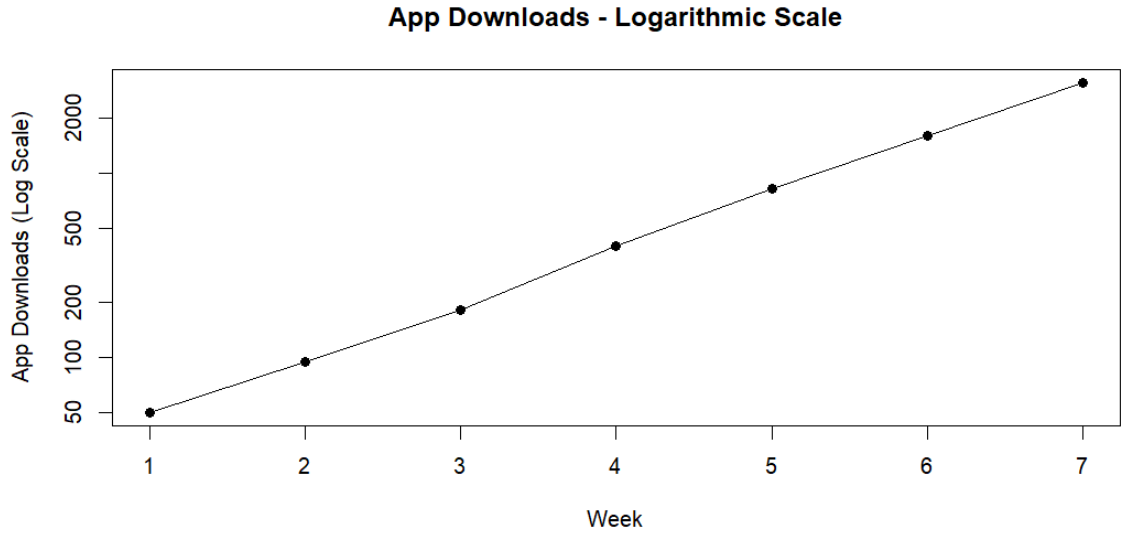
OUTPUT:



2. Modify the program to display the same data using a logarithmic y-axis (plt.yscale('log')).

```
# Data
week <- c(1, 2, 3, 4, 5, 6, 7)
downloads <- c(50, 95, 180, 400, 820, 1600, 3100)
# Logarithmic Y-axis
plot(week, downloads,
    type = "o",
    log = "y",
    main = "App Downloads - Logarithmic Scale",
    xlab = "Week",
    ylab = "App Downloads (Log Scale)",
    pch = 16)
```

OUTPUT:



3. Compare Both Visualizations

Linear Y-axis:

- The Y-axis uses equal intervals such as 500, 1000, 1500, 2000, etc.
- The large values, especially **1600 and 3100**, dominate the graph.
- The difference between smaller values such as **50, 95, and 180** is difficult to observe clearly.
- It is useful when the data values are relatively close to each other.
- It shows the **absolute difference** between values clearly.

Logarithmic Y-axis:

- The Y-axis increases by powers or multiplication factors rather than equal differences.
- Small and large values can be displayed together more clearly.
- The growth pattern from **50 to 3100** is easier to observe.
- It is useful when data covers a **large range of values**.
- It shows **relative or percentage growth** more clearly.

Overall comparison:
The linear graph is easier for understanding exact numerical differences, while the logarithmic graph is better for understanding the rapid growth pattern in this dataset.

4. Why is a Logarithmic Scale Suitable for This Dataset?

A logarithmic scale is suitable because the number of app downloads increases very rapidly over the seven weeks.

The downloads are:

50 → 95 → 180 → 400 → 820 → 1600 → 3100

There is a very large difference between the smallest value (**50**) and the largest value (**3100**).

For example:

- Week 1 = **50 downloads**
- Week 4 = **400 downloads**
- Week 7 = **3100 downloads**

The data increases by approximately multiplicative factors rather than by a constant amount. Therefore, a logarithmic scale compresses the large values and makes the overall growth pattern easier to visualize.

A logarithmic scale is especially useful for **exponential growth, population growth, app downloads, website traffic, financial growth, and scientific measurements** where values can increase by several orders of magnitude.

5. Situation Where a Logarithmic Scale Should NOT Be Used

A logarithmic scale should **not be used when the dataset contains zero or negative values**, because logarithms of zero and negative numbers are undefined.

For example:

Temperature = -10, -5, 0, 5, 10

A logarithmic Y-axis cannot properly represent these values.

It is also not suitable when the main purpose is to compare **exact absolute differences**.

For example, if a teacher wants to compare student marks:

Student A = 40

Student B = 50

Student C = 60

Student D = 70

Student E = 80

A **linear scale** is more appropriate because the difference of 10 marks between students is important and easy to interpret using equal intervals.

Conclusion:
Use a logarithmic scale for **rapidly growing data with a wide range of positive values**. Use a linear scale when **absolute differences are more important or when zero/negative values are present**.

Question 2: Legends and Annotations [Python]

Questions

1. Write a Python program to plot the monthly stock-price trend for all three companies on a single chart with a clear

legend.

2. Annotate (using plt.annotate()) the point of highest price for each company.

Dataset

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")
companyX <- c(410, 430, 460, 480, 520, 500)
companyY <- c(380, 395, 410, 430, 460, 490)
companyZ <- c(450, 460, 440, 470, 510, 530)
# Plot Company X
plot(companyX,
    type = "o",
    pch = 16,
    xaxt = "n",
    ylim = c(350, 550),
```

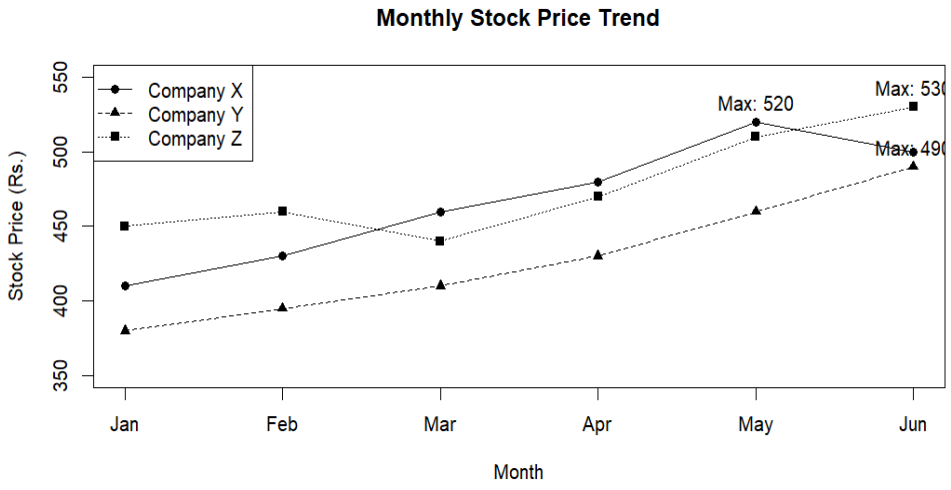
```
    main = "Monthly Stock Price Trend",
    xlab = "Month",
    ylab = "Stock Price (Rs.)")
axis(1, at = 1:6, labels = month)
# Add Company Y and Z
lines(companyY, type = "o", pch = 17, lty = 2)
lines(companyZ, type = "o", pch = 15, lty = 3)
# Add legend
legend("topleft",
       legend = c("Company X", "Company Y", "Company Z"),
       pch = c(16, 17, 15),
       lty = c(1, 2, 3))
# Find highest values
maxX <- which.max(companyX)
maxY <- which.max(companyY)
maxZ <- which.max(companyZ)

# Annotate highest values
text(maxX, companyX[maxX],
    labels = paste("Max:", companyX[maxX]),
    pos = 3)

text(maxY, companyY[maxY],
    labels = paste("Max:", companyY[maxY]),
    pos = 3)

text(maxZ, companyZ[maxZ],
    labels = paste("Max:", companyZ[maxZ]),
    pos = 3)
```

OUTPUT:



3. Role of Legends in Multi-Series Visualizations

A legend helps the viewer identify which line, symbol, or pattern represents each data series. In this graph, three companies are displayed on the same chart, so the legend allows us to distinguish **Company X, Company Y, and Company Z**.

Legends are important because they make multi-series graphs easier to understand and prevent confusion when several datasets are shown together. They also help users compare the trends of different categories without requiring separate graphs.

4. Best Practices for Placing Legends

The legend should be positioned where it does not hide important data points or trends. Common positions include the **top-left, top-right, bottom-left, or bottom-right** of the graph.

If the graph is crowded, the legend can be placed outside the plotting area. The legend should also use the same line styles, symbols, or patterns as the corresponding data series. It should be concise and easy to read.

For example:

```
legend("topleft",
    legend = c("Company X", "Company Y", "Company Z"),
    pch = c(16, 17, 15),
    lty = c(1, 2, 3))
```

The main goal is to choose a location with enough empty space so that the legend **does not obscure the plotted data**.

5. Alternative to a Legend – Direct Labeling

Instead of using a separate legend, we can place the company names directly near their corresponding lines. This technique is called **direct labeling**.

For example:

```
text(6, companyX[6], "Company X", pos = 4)
text(6, companyY[6], "Company Y", pos = 4)
text(6, companyZ[6], "Company Z", pos = 4)
```

Direct labeling has several advantages: it reduces the need to repeatedly look between the graph and legend, makes each line easier to identify, and can make the visualization more intuitive.

However, direct labels should be positioned carefully so that they **do not overlap with each other or with data points**.

Question 3: Mapping Data onto Aesthetics [R]

Dataset

Dataset

Item	Orders	Rating (out of 5)	Category
I1	320	4.2	Electronics

Item	Orders	Rating (out of 5)	Category
I2	540	3.8	Grocery
I3	210	4.6	Fashion
I4	690	4.0	Grocery
I5	150	4.8	Fashion
I6	410	3.5	Electronics

Questions

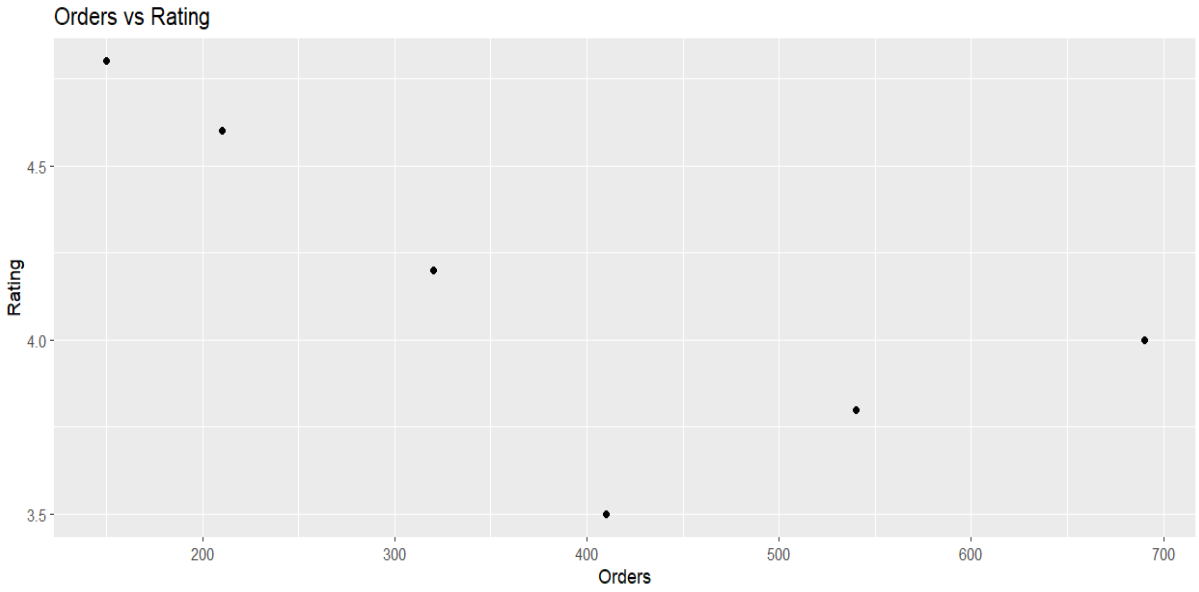
1. Write an R program (using ggplot2) to create a scatter plot with Orders on the x-axis and Rating on the y-axis.

```
# Create dataset
data <- data.frame(
  Item = c("I1", "I2", "I3", "I4", "I5", "I6"),
  Orders = c(320, 540, 210, 690, 150, 410),
  Rating = c(4.2, 3.8, 4.6, 4.0, 4.8, 3.5),
  Category = c("Electronics", "Grocery", "Fashion",
               "Grocery", "Fashion", "Electronics")
)

# Scatter plot
ggplot(data, aes(x = Orders,
                 y = Rating,
                 colour = Category,
                 size = Orders)) +
  geom_point() +
  labs(
    title = "Orders vs Rating by Category",
```

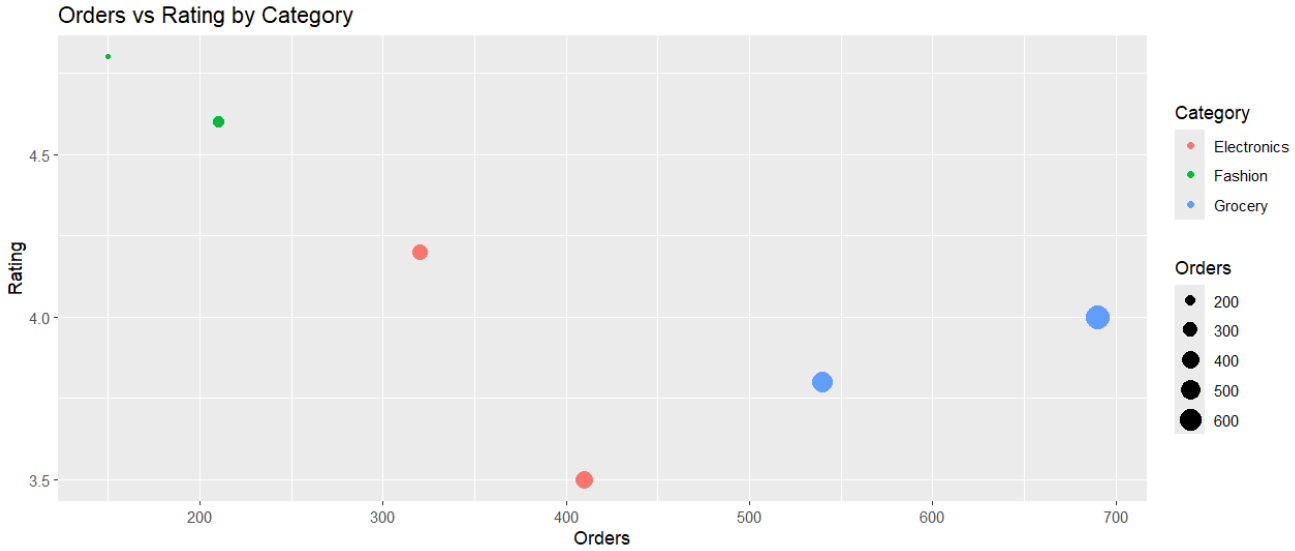
```
  x = "Orders",
  y = "Rating",
  colour = "Category",
  size = "Orders"
)
```

OUTPUT:



2. Map Category using an appropriate aesthetic (e.g. colour) and Orders using another suitable aesthetic (e.g. size).

OUTPUT:



3. Interpretation of the Relationship

From the data, there does not appear to be a strong positive relationship between **Orders** and **Rating**.

- **I4** has the highest number of orders (**690**) but a rating of only **4.0**.
- **I5** has only **150 orders** but the highest rating (**4.8**).
- **I6** has **410 orders** and the lowest rating (**3.5**).
- **I3** has **210 orders** with a high rating of **4.6**.

Therefore, higher orders do **not necessarily mean higher ratings**.

The relationship appears to be **weak or inconsistent**, based on the six observations.

4. Type of Scale for Each Variable

Variable	Type of Scale	Explanation
Item	Nominal	Item IDs are categories/names without numerical meaning
Orders	Continuous	Numerical quantity that can be measured
Rating	Ordinal	Ratings have an ordered scale from low to high
Category	Nominal	Categories such as Electronics, Grocery and Fashion have no inherent order

Note: In many practical statistical analyses, a numerical rating such as 4.2, 3.8, etc. may be treated as a **continuous variable**. But because the question asks for scale types, the rating is commonly classified as **ordinal** when interpreted as a rating scale.

5. Suggest one alternative visualization that better represents the same data.

```
ggplot(data, aes(x = Item, y = Orders, fill = Category)) +
 geom_bar(stat = "identity") +
 labs(
  title = "Orders by Item",
  x = "Item",
  y = "Number of Orders",
  fill = "Category"
 )
```

Question 4: Highlighting with Color [R]

Dataset

Questions

1. Write an R program to create a bar chart highlighting the expense category with the highest monthly spend.

```
# Create data
category <- c("Rent", "Groceries", "Travel", "Utilities", "Entertainment")
spend <- c(15000, 8000, 4000, 3000, 2500)

# Find the highest spending category
highest <- which.max(spend)

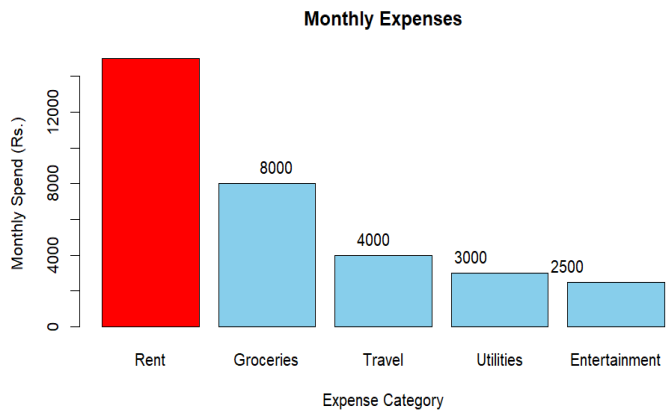
# Assign colors
colors <- rep("skyblue", length(spend))
colors[highest] <- "red"

# Create bar chart
barplot(spend,
        names.arg = category,
        col = colors,
        main = "Monthly Expenses",
        xlab = "Expense Category",
        ylab = "Monthly Spend (Rs.)")

# Display values
text(seq_along(spend), spend,
    labels = spend,
    pos = 3)

# Highlight highest category
text(highest, spend[highest],
    labels = "Highest",
    pos = 3)
```

OUTPUT:



2. How Highlighting Improves Data Interpretation

Highlighting draws attention to the most important data point. In this chart, the highest expense is shown in a different color, so the viewer can immediately identify it without comparing every bar.

It helps to:

- Draw attention to important information.
- Make key values easier to identify.
- Improve visual understanding.
- Allow quick comparison between the highlighted value and other values.

3. Highest-Spending Category

The **highest-spending category is Rent**.

Monthly Rent = ₹15,000

It is highlighted in **red** in the bar chart.

4. Effect of Excessive Use of Colors

Using too many colors can make a visualization confusing and difficult to interpret.

For example, if every expense category has a different bright color, the viewer may not know which color represents the important category. Excessive colors can:

- Distract the viewer.
- Reduce visual clarity.
- Make comparisons difficult.
- Make the chart look cluttered.
- Reduce the effectiveness of highlighting.

Therefore, color should be used **selectively**, mainly to emphasize important information.

5. Alternative Highlighting Technique

An alternative to color is **using a different pattern or border**.

For example, the highest bar can be given a thicker border:

```
barplot(spend,
        names.arg = category,
        col = "skyblue",
        border = ifelse(seq_along(spend) == highest, "black", "white"),
```

```
lwd = ifelse(seq_along(spend) == highest, 4, 1),
main = "Monthly Expenses",
xlab = "Expense Category",
ylab = "Monthly Spend (Rs.)")
```

Other alternatives include:

- Adding a label such as **"Highest"**
- Using a different bar pattern
- Making the bar wider
- Adding an arrow or annotation
- Using a bold outline

Best simple alternative: Add an annotation such as **"Highest – ₹15,000"** above the Rent bar.

Question 5: Integrated Visualization [Python]

Dataset

Country	GDP (₹ Trillion)	Literacy Rate (%)	Population (Million)	Continent
C1	12.5	92	140	Asia
C2	8.2	88	65	Europe
C3	21.0	95	330	N. America
C4	3.1	75	210	S. America
C5	5.6	80	95	Africa

Questions

Questions

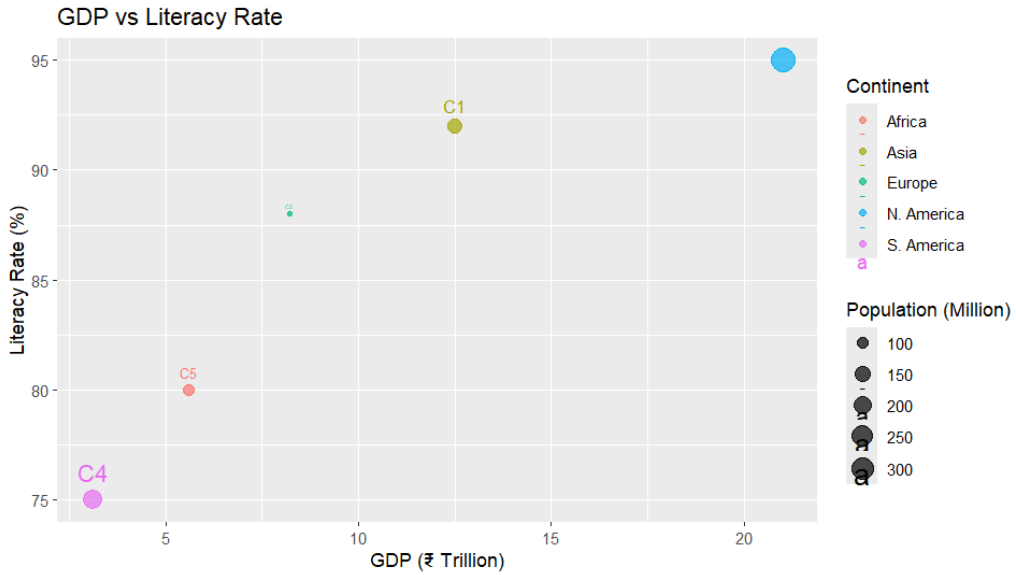
1. Write a Python program to create a bubble chart using appropriate aesthetic mappings.

```
library(ggplot2)

data <- data.frame(
  Country = c("C1", "C2", "C3", "C4", "C5"),
  GDP = c(12.5, 8.2, 21.0, 3.1, 5.6),
  Literacy = c(92, 88, 95, 75, 80),
  Population = c(140, 65, 330, 210, 95),
  Continent = c("Asia", "Europe", "N. America",
                "S. America", "Africa")
)
```

```
ggplot(data, aes(x = GDP,
                 y = Literacy,
                 size = Population,
                 color = Continent)) +
  geom_point(alpha = 0.7) +
  geom_text(aes(label = Country), vjust = -1) +
  labs(
    title = "GDP vs Literacy Rate",
    x = "GDP (₹ Trillion)",
    y = "Literacy Rate (%)",
    size = "Population (Million)",
    color = "Continent"
  )
```

OUTPUT:



2. Suitable Aesthetics

Variable	Visualization Aesthetic
GDP	X-axis position
Literacy Rate	Y-axis position
Population	Bubble size
Continent	Color
Country	Label

This allows four different variables to be represented in a single visualization.

3. Interpretation of GDP and Literacy Rate

There appears to be a **positive relationship** between GDP and literacy rate in this dataset.

Examples:

- **C3:** GDP = ₹21.0 trillion and literacy = **95%**
- **C1:** GDP = ₹12.5 trillion and literacy = **92%**
- **C2:** GDP = ₹8.2 trillion and literacy = **88%**
- **C5:** GDP = ₹5.6 trillion and literacy = **80%**
- **C4:** GDP = ₹3.1 trillion and literacy = **75%**

Generally, countries with higher GDP have higher literacy rates in this small dataset.

However, this is only an **observed relationship** from five countries and does not prove that higher GDP causes higher literacy.

4. Continuous and Categorical Variables

Continuous Variables

- **GDP** – Continuous
- **Literacy Rate** – Continuous
- **Population** – Continuous

These variables have numerical values and can be measured quantitatively.

Categorical Variables

- **Country** – Categorical/nominal
- **Continent** – Categorical/nominal

These variables represent groups or categories rather than numerical measurements.

5. Improvement for Better Readability

One good improvement is to **add country labels directly to each bubble**.

For example:

plt.annotate("C1", (12.5, 92))

Labels make it easier to identify individual countries without repeatedly referring to a separate table.

Other improvements include:

- Use transparency (alpha) to avoid overlapping bubbles.
- Add a clear legend for continents.
- Use an appropriate bubble-size scale.
- Add grid lines.
- Use a larger figure size.
- Keep the number of colors limited and meaningful.

Best improvement: Use **direct country labels + a clear continent legend**, because the chart contains only five countries.